Advanced Winter Olympic Medal Analysis Using Pig Latin

Assignment Two

Assignment name: Advanced Winter Olympic Medal Analysis Using Pig Latin

Course code: COSC2633/2637

Weighting: 25/100

Due date: 27/09 at 23:59

1. Course Learning Outcomes Assessed

This assessment supports the following learning outcomes:

- CLO 1: model and implement efficient big data solutions for various application areas using appropriately selected algorithms and data structures.

- CLO 2: analyse methods and algorithms, to compare and evaluate them with respect to time and space requirements and make appropriate design choices when solving real-world problems.
- CLO 3: motivate and explain trade-offs in big data processing technique design and analysis in written and oral form.
- CLO 4: explain the Big Data Fundamentals, including the evolution of Big Data, the characteristics of Big Data and the challenges introduced.
- CLO 6: apply the novel architectures and platforms introduced for Big data, i.e., Hadoop, MapReduce and Spark.

2. Overview of Assessment

Write Apache Pig scripts which give you a chance to develop a basic understanding of principles when handling queries on large data stored on HDFS.

3. Assessment Tasks

Follow the steps below to get started on this assessment task:

- **A2 specification**
- **countries.csv**
- **games.csv**
- **medal_table.csv**
- **population.csv**
- **A2_GroupLeaderStudentID_GroupNumber_UGorPG_worklog.docx**
- **Apache Pig UDF Guide: https://pig.apache.org/docs/r0.17.0/udf.html#jython-udfs**

4. Group Assessment

Groups must be formed by **2 or 3 students** who are enrolled at the same level (either all Undergraduate COSC2633 or all Postgraduate COSC2637). All members are expected to contribute equally. Unless a contribution dispute is reported (and supported by the work log), all group members will receive the same grade.

Individual work is permitted but not encouraged. Please note that no bonus marks will be awarded for completing the work individually.

- **Navigate to People in Canvas, then select the Groups tab. Scroll past the A1 groups until you reach the A2 group section.**
- **Choose an empty A2 group and join it together with your group members.**
- **Please do not use your A1 group for this assessment. You may keep the same team members, but you must join a A2 group.**
- **You are not permitted to create new groups. You are not permitted to rename the groups.** This restriction is in place because student-created groups are not linked to the grading system, which means they cannot be identified by the system during marking.
- Canvas groups are intended only for students completing the assignment in groups of two or three. Students completing the assignment individually do not need to join a Canvas group.

Please note that it is each student's responsibility to ensure they have successfully joined the Canvas group before commencing work on Assessment 2.

Group membership should be confirmed by individual students before work begins. The team leader's role is generally to verify the group composition before and after submission. However, this does not remove each team member's responsibility to ensure that they are correctly registered in the group throughout the assessment period.

Detailed requirements and instructions can be found in the assignment specification document.

5. Submission Instructions

Detailed requirements and instructions can be found in the assignment specification document.

Students who choose to complete the assessment collaboratively but fail to correctly form their group before the submission deadline will receive a 10% deduction (2.5 marks).

Final Submission Warning

Please verify your submission carefully before the deadline. Ensuring that the correct files have been submitted is **the responsibility of the entire group**, not just the group leader. All group members are expected to check that the submission is complete, correct, and submitted before the deadline.

Groups will be locked 72 hours before the deadline. You must finalise and verify your group membership **before submission** and only submit after groups are locked. **Please do not submit your assignment until the groups have been finalised and locked.**

- You must submit 3 files: **the zip file, the doc file, and the pdf file** for Turnitin plagiarism check.
- Failure to submit the required pdf/doc file separately from the zip file will result in a 10% penalty (2.5 marks).
- Submissions that do not include the required PDF or Word document will incur a 10% penalty for each missing file (2.5 marks per file).
- Submissions where the PDF **fails the Turnitin plagiarism check**, **will not be graded**.
- Please verify your submission carefully before the deadline. If you submit the **wrong file**, a **corrupted zip file**, or **a submission that does not comply with the required file structure**, any resubmission made after the deadline will be subject to the standard late submission penalty.

- Late submission penalties are applied automatically by the system based on the assessment due date and the time of submission. **The penalty is determined by when you submit your work, not when you complete it.** Therefore, even if you completed the assessment before the due date, a late submission would still incur the applicable late penalty.
- Work must run on the **RMIT-provided EMR.**
- **Please note that markers will not modify or adjust submitted code during marking.**

6. Assessment Criteria

The marking guides are included in the assignment specification.

Marks will be awarded based on the quality of your design and implementation, rather than solely on producing the correct output. Solutions should demonstrate a sound understanding of Pig Latin principles.

Criteria	Ratings	Pts
Pig Latin Implementation (17 marks) Undergraduate (COSC2633) ========== 1. Correct data loading and joins (3 marks) 2. Medal efficiency calculation (4 marks) Correctly computes total_medals and medals_per_million. 3. Region filtering using parameter (2 marks) 4. Ranking logic (3 marks)		17 pts

Criteria	Ratings	Pts
5. Output format (2 marks)		
6. Pig workflow quality (3 marks)		
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Postgraduate (COSC2637)		
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1. Correct data loading and joins (3 marks)		
2. Top 3 by gold medals (1 marks)		

Criteria	Ratings	Pts
3. Top 3 by total medals (2 marks) 4. Host city and country names (2 marks) 5. Year filtering using parameter (3 marks) 6. Output format (3 marks) 7. Pig workflow quality (3 marks)		
Python UDF (5 marks) Undergraduate (COSC2633) ==================================		5 pts

Criteria	Ratings	Pts
Correct ranking and reusable implementation. ========== Postgraduate (COSC2637) ========== Produces required pretty-print output exactly.		
Parameterisation (3 marks) Undergraduate (COSC2633) ==========		3 pts

Criteria	Ratings	Pts
REGION parameter fully implemented. ========== Postgraduate (COSC2637) ========== Uses START_YEAR and END_YEAR correctly.		
Common Deductions (format and submission) −2.5 marks: Missing PDF (Turnitin). The PDF file for Turnitin plagiarism check must be submitted separately from the ZIP file. If the PDF is included only inside the ZIP or is missing entirely, a 10% penalty will apply.		0 pts

Criteria	Ratings	Pts
−2.5 marks: Missing DOC work log −2.5 marks: Incorrect ZIP structure −2.5 marks: Incorrect file names (Pig script or UDF) −2.5 marks: HDFS input dir or/and HDF output dir is/are incorrectly specified. −1 mark: Pig script is incorrectly named.		
Functional Requirements −2.5 marks: The code is not well written using a good coding style. −2.5 marks: The codes and scripts do not come with concise and clear comments to explain the logical flow of the program.		0 pts
		Total Points: 25